Homework 2

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Problem 1

1. Exponent rule
$$\int_0^1 a^y \, \mathrm{d}y = \left.\frac{a^y}{\ln a}\right|_0^1 = \frac{a-1}{\ln a}.$$

2. U-substitution, $u = ay$, $du = adx$, $v = -ay$, $dv = -vdx$.
$$\int_0^x \cosh y \, \mathrm{d}y = \int_0^x \frac{e^{ax} + e^{-ax}}{2} \, \mathrm{d}y = \frac{1}{2} \left(\int_0^x e^{ay} \, \mathrm{d}y + \int_0^x e^{-ay} \, \mathrm{d}y \right)$$
$$= \frac{1}{2a} \left(\int_0^{ax} e^u \, \mathrm{d}u - \int_0^{-ax} e^v \, \mathrm{d}v \right) = \frac{1}{2a} \left. e^u \right|_0^{ax} - \left. e^v \right|_0^{-ax} = \frac{1}{2a} \left(e^{ax} - 1 - e^{-ax} + 1 \right)$$
$$= \frac{1}{a} \frac{e^{ax} - e^{-ax}}{2} = \frac{1}{a} \sinh(ax).$$

3. Derivative of $\ln x$, u substitution $u = ay + b$, $du = 1/a \, dx$.
$$\int_0^x \frac{1}{ay+b} \, \mathrm{d}y = \int_0^x (ax+b)^{-1} \, \mathrm{d}y = \int_0^x \frac{1}{a} u^{-1} \, \mathrm{d}u = \left. \frac{1}{a} \ln(ay+b) \right|_0^x = \frac{1}{a} \left(\ln(ax+b) - \ln b \right).$$

4. IBP, let $g = y^2,\ g' = 2y,\ f' = \cos(2y),\ f = 1/2 \cos(2y)$.
$$\int_0^x y^2 \cos(2y) \, \mathrm{d}y = \left. \frac{y^2}{2} \sin(2y) \right|_0^x - \int_0^x \frac{2y}{2} \sin(2y) \, \mathrm{d}y = \frac{x^2}{2} \sin(2x) - \int_0^x y \sin(2y) \, \mathrm{d}y.$$
IBP, let $g = y,\ g' = 1,\ f' = \sin(2y),\ f = -1/2 \cos(2y)$.
$$= \frac{x^2}{2} \sin(2x) + \left. \frac{y}{2} \cos(2y) \right|_0^x - \int_0^x \frac{1}{2} \cos(2y) \, \mathrm{d}y = \frac{x^2}{2} \sin(2x) + \frac{x}{2} \cos(2x) - \left. \frac{1}{4} \sin(2y) \right|_0^x$$
$$= \frac{1}{2} \left(x^2 \sin(2x) + x \cos(2x) - \frac{1}{2} \sin(2x) \right).$$

5. U-substitution, let $u = -x^4$, $du = -4x^3 dx$.
$$\int_0^z x^3 e^{-x^4} \, \mathrm{d}x = \int_0^{-z^4} -\frac{1}{4} e^u \, \mathrm{d}u = \left. -\frac{1}{4} e^u \right|_0^{-z^4} = \frac{1}{4} \left(-e^{-z^4} + 1 \right).$$

6. Starting with a proof that $\cosh^2 x - \sinh^2 x = 1$.

$$\left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 = \left(\frac{e^{2x} + 2 + e^{-2x}}{4}\right) - \left(\frac{e^{2x} - 2 + e^{-2x}}{4}\right) = \frac{4}{4} = 1.$$

U-substitution, $u = \sinh x$, $du = \cosh x$.

$$\int_0^z \cosh^3 x \, \mathrm{d}x = \int_0^z \left(1 + \sinh^2 x\right) \cosh x \, \mathrm{d}x = \int_0^{\sinh z} 1 + u^2 \, \mathrm{d}u = \left(u + \frac{1}{3}u^3\right)\Big|_0^{\sinh z}$$

$$= \sinh z + \frac{1}{3}\sinh^3 z.$$

7. Trig substitution, let $x = a\sin\theta$. Then $dx = a\cos\theta d\theta$. Also if $x = z$ then $\theta = \arcsin\frac{z}{a}$, and if $x = 0$ then $\theta = 0$.

$$\int_0^z \frac{1}{\sqrt{a^2 - x^2}} \, \mathrm{d}x = \int_0^{\arcsin z/a} \frac{a\cos\theta}{\sqrt{a^2\left(1 - \sin^2\theta\right)}} \, \mathrm{d}\theta = \int_0^{\arcsin z/a} \frac{a\cos\theta}{a\sqrt{\cos^2\theta}} \, \mathrm{d}\theta$$

$$= \int_0^{\arcsin z/a} \frac{a\cos\theta}{a\cos\theta} \, \mathrm{d}\theta = \int_0^{\arcsin z/a} \mathrm{d}\theta = \arcsin\frac{z}{a}.$$

This works because we take $|z| < a$, so $|z/a| < 1$, and arcsin is well-defined on this interval.

Problem 2

Let $g = \ln x$, $g' = 1/x$, $f' = 1$, $f = x$. Then

$$\int \ln x \, \mathrm{d}x = x\ln x - \int \frac{x}{x} \, \mathrm{d}x = x\ln x - \int \mathrm{d}x = x\ln x - x + C.$$

Problem 3

Isn't this basically just $\Gamma(n+1)$? First, we show $F(n) = nF(n-1)$ for $n \geq 1$. Let $g = x^n$ and $f' = e^{-x}$. Then $g' = nx^{n-1}$ and $f = -e^{-x}$. We have

$$F(n) = \int_0^\infty x^n e^{-x} \, \mathrm{d}x = -x^n e^{-x}\Big|_0^\infty + \int_0^\infty nx^{n-1}e^{-x} \, \mathrm{d}x = \left(\lim_{x\to\infty} \frac{-x^n}{e^x}\right) + nF(n-1).$$

It suffices to show the limit is 0. Note that if $n - 1 \geq 1$, we have

$$\lim_{x\to\infty} \frac{-x^n}{e^x} = -\frac{\infty}{\infty}.$$

Therefore, we can apply L'Hopital's rule until the exponent is 0. We then have

$$\lim_{x\to\infty} \frac{-x^n}{e^x} = \lim_{x\to\infty} \frac{-n!}{e^x} = 0$$

because n is finite, so $n!$ is finite.

Next, we will compute $F(0)$.

$$F(0) = \int_0^\infty e^{-x} \, \mathrm{d}x = -e^{-x}\Big|_0^\infty = 0 - (-1) = 1.$$

Therefore, $F(0) = 1$.

Now we will show $F(n) = n!$ for $n \in \mathbb{N}$ using the method of induction. Since $F(n) = nF(n-1)$, and $F(0) = 1$, we have

$$F(1) = 1 \cdot F(0) = 1.$$

Therefore, $F(1) = 1 = 1!$. Now suppose that $F(n) = n!$ for some arbitrary $n \in \mathbb{N}$. It follows that

$$F(n+1) = (n+1)F(n) = (n+1)n! = (n+1)!.$$

Therefore, $F(n+1) = (n+1)!$, and the statement is proven.

The integral defines $0! = 1$ because $F(0) = 1$.

Problem 4

1. Let $u = \sqrt{a}x$. Then

$$I_0(a) = \int_{-\infty}^{\infty} e^{-ax^2}\,\mathrm{d}x = \frac{1}{\sqrt{a}}\int_{-\infty}^{\infty} e^{-u^2}\,\mathrm{d}u = \frac{1}{\sqrt{a}} I_0(1).$$

2. Since n is odd, let $n = 2m+1$ for some $m \in \mathbb{N}$. We have

$$I_n(a) = \int_{-\infty}^{\infty} x^{2m+1} e^{-ax^2}\,\mathrm{d}x = \int_{-\infty}^{\infty} x\left(\frac{2a}{2a}x^2\right)^m e^{-ax^2}\,\mathrm{d}x.$$

Let $u = ax^2$. Then $du = 2axdx$. Note that applying this u-substitution will cause both bounds to now be ∞ instead of $\pm\infty$, so I will break the integral into two parts to make stuff work. We have

$$= \frac{1}{(2a)^{m+1}}\left(\int_0^{\infty} u^m e^{-u}\,\mathrm{d}u + \int_{-\infty}^{0} u^m e^{-u}\,\mathrm{d}u\right).$$

$$\begin{aligned}
\int_{-\infty}^{\infty} x\left(\frac{2a}{2a}x^2\right)^m e^{-ax^2}\,\mathrm{d}x &= \int_{-\infty}^{0} x\left(\frac{2a}{2a}x^2\right)^m e^{-ax^2}\,\mathrm{d}x + \int_{-\infty}^{0} x\left(\frac{2a}{2a}x^2\right)^m e^{-ax^2}\,\mathrm{d}x \\
&= \int_{-\infty}^{0} \frac{1}{(2a)^{m+1}} u^m e^{-u}\,\mathrm{d}u + \int_{0}^{\infty} \frac{1}{(2a)^{m+1}} u^m e^{-u}\,\mathrm{d}u \\
&= \frac{1}{(2a)^{m+1}}\left(-\int_{0}^{\infty} u^m e^{-u}\,\mathrm{d}u + \int_{0}^{\infty} u^m e^{-u}\,\mathrm{d}u\right).
\end{aligned}$$

Since both integrals are now the same, they cancel out, and $I_n(a) = 0$. Interestingly, note both of these integrals are $F(m)$ as defined in problem 3.

3. We use proof by the method of induction. Let $p = 1$. Then

$$(-1)\frac{\mathrm{d}I_0(a)}{\mathrm{d}a} = -\int_{-\infty}^{\infty} \frac{\mathrm{d}}{\mathrm{d}a} e^{-ax^2}\,\mathrm{d}x = -\int_{-\infty}^{\infty} \left(-x^2\right) e^{-ax^2}\,\mathrm{d}x = \int_{-\infty}^{\infty} x^2 e^{-ax^2}\,\mathrm{d} = I_2(a).$$

Now suppose the statement holds for some arbitrary $p \in \mathbb{N}$. We have

$$(-1)^{p+1}\frac{\mathrm{d}^{p+1}I_0(a)}{\mathrm{d}a^{p+1}} = -\int_{-\infty}^{\infty} \frac{\mathrm{d}}{\mathrm{d}a} x^{2p} e^{-ax^2}\,\mathrm{d}x = -\int_{-\infty}^{\infty} x^{2p}\left(-x^2\right) e^{-ax^2}\,\mathrm{d}x$$

$$= \int_{-\infty}^{\infty} x^{2(p+1)} e^{-ax^2}\,\mathrm{d}x = I_{2(p+1)}(a).$$

Therefore,

$$(-1)^p \frac{\mathrm{d}^p I_0(a)}{\mathrm{d}a^p} = I_{2p}(a).$$

4. The problem does not ask for proof, only for the pattern.

$$I_0(a) = \sqrt{\frac{\pi}{a}}.$$

We know this from part 1 and commentary in the problem set.

$$I_2(a) = -\frac{\mathrm{d}I_0(a)}{\mathrm{d}a} = \frac{\mathrm{d}}{\mathrm{d}a}\sqrt{\pi}a^{-1/2} = \frac{1}{2}\sqrt{\frac{\pi}{a^3}}.$$

$$I_4(a) = -\frac{\mathrm{d}I_2(a)}{\mathrm{d}a} = -\frac{\mathrm{d}}{\mathrm{d}a}\frac{1}{2}\sqrt{\pi}a^{-3/2} = \frac{3}{2}\cdot\frac{1}{2}\sqrt{\frac{\pi}{a^5}}.$$

In general, the formula is

$$I_{2n}(a) = \frac{1}{2^n}\left(\prod_{i=0}^{n-1} 2i+1\right)\sqrt{\frac{\pi}{a^{2n+1}}} = \frac{(2n)!}{2^{2n}n!}\sqrt{\frac{\pi}{a^{2n+1}}}.$$

This second representation uses this identity.

Problem 5

1. The argument is exactly the same, so I will show less work. We will prove by induction. Let $p = 1$. then

$$-\frac{\mathrm{d}\widetilde{I}_1(a)}{\mathrm{d}a} = -\int_0^{\infty} \frac{\mathrm{d}}{\mathrm{d}a} xe^{-ax^2}\,\mathrm{d}x = \int_0^{\infty} x^3 e^{-ax^2}\,\mathrm{d}x = \widetilde{I}_3(a) = \widetilde{I}_{2+1}(a).$$

Now suppose the statement holds for $p \in \mathbb{N}$ arbitary, and consider $p+1$. We have

$$-\frac{\mathrm{d}^{p+1}\widetilde{I}_1(a)}{\mathrm{d}a^{p+1}} = -\int_0^{\infty} \frac{\mathrm{d}}{\mathrm{d}a} x^{2p+1} e^{-ax^2}\,\mathrm{d}x = \int_0^{\infty} x^{2p+3} e^{-ax^2}\,\mathrm{d}x = \int_0^{\infty} x^{2(p+1)+1} e^{-ax^2}\,\mathrm{d}x$$

$$= \widetilde{I}_{2(p+1)+1}(a).$$

By the method of induction, the theorem is proven.

2. Let $u = ax^2$. Then $du = 2axdx$. Recall the definition of $F(n)$ from problem 3, and that $F(0) = 1$. We have

$$\widetilde{I}_1(a) = \int_0^{\infty} xe^{-ax^2}\,\mathrm{d}x = \frac{1}{2a}\int_0^{\infty} e^{-u}\,\mathrm{d}u = \frac{1}{2a}F(0) = \frac{1}{2a}.$$